

Finite generation of canonical rings

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Outline of the talk

1 Introduction

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- 2 The MMP for 3-folds

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 - The main Theorem
 - Curves
 - Surfaces
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- K_X denotes (a choice of) the canonical divisor i.e. a formal linear combination of codimension 1 subvarieties of X given by the zeroes of a section $s \neq 0$ of ω_X .
- We would like to use ω_X (or equivalently K_X) to understand the geometry of X .

Pluricanonical maps

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$$\varphi_{\omega_X^{\otimes m}} : X \dashrightarrow \mathbb{P}^N$$

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- If X and X' are birational, then $\forall m \geq 0$

$$H^0(X, \omega_X^{\otimes m}) \cong H^0(X', \omega_{X'}^{\otimes m}).$$

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- One would like to understand the structure of this ring and to use its features to classify complex projective varieties.

Finite generation

Today I would like to discuss the following result:

Theorem (Birkar-Cascini-Hacon-M^cKernan and Siu)

Let X be a smooth projective variety, then $R(X)$ is finitely generated.

The two proofs are independent. Our proof is algebraic and uses the ideas of the MMP (minimal model program). Siu's proof is analytic and requires X to be of general type.

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These form a $3g - 3$ -dimensional family.
 K_X is ample and $\varphi_{\omega_X^{\otimes 3}}$ determines an embedding.
- $\omega_X^{\otimes 3} = \mathcal{O}_{\mathbb{P}^N}(1)|_X$ so that $R(X)$ is finitely generated (by Serre-Vanishing).

Birational geometry of surfaces

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- The morphism ν may be viewed as a surgery that replaces the point x by a rational curve $E \cong \mathbb{P}^1 = \mathbb{P}(T_x X)$.
- The exceptional curve E is called a -1 curve since

$$\omega_{\tilde{X}} \cdot E = \deg(\omega_{\tilde{X}}|_E) = -1.$$

Birational geometry of surfaces II

- By a Theorem of Castelnuovo, one may reverse this procedure i.e. given any surface X and any -1 curve $E \subset X$, there exists a morphism $\nu : X \rightarrow X_1$ such that $X = \text{Bl}_x(X_1)$ for some point $x \in X_1$.

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- Note that $\rho(X_1) = \rho(X) - 1 \geq 0$ and so this procedure can be repeated at most finitely many times.
- A minimal surface is a surface that contains no -1 curves.
- By Castelnuovo's Theorem, given any surface X there exists a birational morphism to a minimal surface $X \rightarrow X_{\min}$ (given by finitely many point blow downs).

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- If $\kappa(X) \geq 0$, then the minimal model $X \rightarrow X_{min}$ is unique and $K_{X_{min}}$ is nef. This means that for any curve $C \subset X_{min}$ we have

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- In fact one can show that $\omega_{X_{\min}}$ is semiample i.e. there is an integer $m > 0$ such that the sections of $H^0(\omega_{X_{\min}}^{\otimes m})$ have no common zeroes.

Birational geometry of surfaces III

- Therefore, $\varphi_{\omega_{X_{min}}^{\otimes m}} : X_{min} \rightarrow \mathbb{P}^N$ is a morphism,
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- Conclusion: After performing a finite sequence of geometrically meaningful operations (contracting -1 curves) we obtain a birational model on which ω is positive (nef and semiample).

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- Conclusion: After performing a finite sequence of geometrically meaningful operations (contracting -1 curves) we obtain a birational model on which ω is positive (nef and semiample).
- Finite generation of $R(X)$ then follows from the finite generation of $R(\mathcal{O}_Z(1))$.

Birational geometry of surfaces IV

More precisely we have the following:

- If $\kappa(X) = 0$, then $12K_{X_{min}} = 0$ so that
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- If $\kappa(X) = 1$, then $\omega_{X_{min}}^{\otimes 12}$ defines a morphism $f : X_{min} \rightarrow C$ whose general fiber is an elliptic curve. We have $\omega_{X_{min}}^{\otimes 12} = f^*H$ where H is an ample line bundle on C .

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- If $\kappa(X) = 2$, then $\omega_{X_{min}}^{\otimes 5}$ defines a morphism $\varphi : X_{min} \rightarrow X_{can} \subset \mathbb{P}^N$ where $X_{can} = \text{Proj}(R(X))$ is the canonical model of X . The map $X \rightarrow X_{can}$ is birational and X_{can} has finitely many mildly singular points (rational double points).

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 - Threefolds
 - Running the minimal model program
- 3 Higher dimensional MMP

MMP for 3-folds

- By work of Mori, Kawamata, Kollár, Reid, Shokurov and others it is possible to generalize this picture to 3-folds:

Theorem

Assume $\dim(X) = 3$. If $\kappa(X) < 0$, then X is covered by rational curves and $R(X) \cong \mathbb{C}$.

If $\kappa(X) \geq 0$, then there is a birational map $X \dashrightarrow X_{\min}$ such that $\omega_{X_{\min}}$ is nef (i.e. $\omega_{X_{\min}} \cdot C \geq 0$ for any curve $C \subset X_{\min}$).

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- More precisely, if $\kappa(X) \geq 0$, there is a sequence of flips and divisorial contractions

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- It is known that $\omega_{X_{\min}}$ is semiample and so $R(X)$ is finitely generated.

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- Flips are a new operation that we will discuss below.
- The minimal model X_{min} is not unique.
- The varieties X_i have mild singularities.

Running a MMP

- A Minimal Model program is a sequence of geometric operations $X_i \dashrightarrow X_{i+1}$ that starting from a projective variety X produces a minimal model (if $\kappa(X) \geq 0$) or a covering family of rational curves (if $\kappa(X) < 0$).

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- We start with a projective variety X with mild singularities.
- If ω_X is nef we are done ($X = X_{min}$).
- If ω_X is not nef, then consider the cone of effective curves

$$\overline{NE}(X) = \left\{ \sum c_i C_i \mid c_i \in \mathbb{R} \right\} / \equiv$$

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- By the Cone Theorem, there is a negative extremal ray $R = \mathbb{R}^+[C]$ where $C \subset X$ is a curve such that $\omega_X \cdot C < 0$ and there is a morphism $\text{Cont}_R : X \rightarrow Z$ such that a curve $D \subset X$ is contracted if and only if $[D] \in R^+[C]$.

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- If $\dim Z = \dim X$ and Cont_R is small (i.e. $\dim \text{Exc}(X \rightarrow Z) \leq \dim X - 2$) then Z does not have mild singularities. In particular $\omega_Z \cdot C$ does not always make sense.

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- Instead of replacing X by Z , we will replace X by its flip X^+ .

Flips

- The flip of $X \rightarrow Z$ is a small birational morphism $X^+ \rightarrow Z$ such that
 - 1) $X \dashrightarrow X^+$ is an isomorphism on the complement of $\text{Exc}(X \rightarrow Z)$,
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 - 3) If $C \subset \text{Exc}(X^+ \rightarrow Z)$, then $\omega_{X^+} \cdot C > 0$.
- In other words, the flip $X \dashrightarrow X^+$ is a surgery that replaces ω_X negative curves by ω_{X^+} positive curves.

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- The flip of $X \rightarrow Z$ is a small birational morphism $X^+ \rightarrow Z$ such that
 - 1) $X \dashrightarrow X^+$ is an isomorphism on the complement of $\text{Exc}(X \rightarrow Z)$,
 - 2) X^+ has mild singularities, and
 - 3) If $C \subset \text{Exc}(X^+ \rightarrow Z)$, then $\omega_{X^+} \cdot C > 0$.
- In other words, the flip $X \dashrightarrow X^+$ is a surgery that replaces ω_X negative curves by ω_{X^+} positive curves.
- If the flip of $X \rightarrow Z$ exists, it is unique and given by $\text{Proj}_Z R(X)$. So that to construct the flip we must show that $R(X)$ is finitely generated over Z (i.e. if $Z = \text{Spec} A$, $R(X)$ is a finitely generated A -algebra).

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- In dimension 3, this is relatively easy to do.

Outline of the talk

- 1 Introduction
- 2 The MMP for 3-folds
- 3 Higher dimensional MMP
 - Reduction to log pairs of general type
 - Idea of the proof

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- The proof proceeds by induction on the dimension of X . We relate ω_X (equivalently K_X) to its restriction to a divisor S (i.e. a codimension 1 subvariety which typically belongs to $|mK_X|$).
- This is achieved by the adjunction formula which has the form $(K_X + S)|_S = K_S + \Delta_S$ for some $\mathbb{Q}$ -divisor Δ_S .

A Theorem of Fujino-Mori

Mori and Fujino have shown that:

Theorem

Let (X, Δ) be any KLT $\mathbb{Q}$ -factorial pair with $\Delta \in \text{Div}_{\mathbb{Q}}(X)$. Then there exists a KLT $\mathbb{Q}$ -factorial pair (Y, Γ) of general type with $\Gamma \in \text{Div}_{\mathbb{Q}}(Y)$ such that

$$R(K_X + \Delta)^{(m)} \cong R(K_Y + \Gamma)^{(m)}$$

for any sufficiently divisible integer $m > 0$.

Recall that $K_Y + \Gamma$ is of general type if $\text{tr.deg.}_{\mathbb{C}} R(K_Y + \Gamma)^{(m)} - 1 = \dim Y$.

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- If (X, Δ) is KLT and $K_X + \Delta$ is nef and of general type, then by the Base Point Free Theorem of Kawamata and Shokurov, $K_X + \Delta$ is semiample.
- As we have mentioned before, this implies that $R(K_X + \Delta)$ is finitely generated.
- Therefore, to show that $R(X)$ is finitely generated (for any smooth complex projective variety), it suffices to show that minimal models exist for KLT pairs of general type.

Existence of minimal models for KLT pairs of general type.

Theorem (Birkar-Cascini-Hacon-M^cKernan)

Let (X, Δ) be any KLT pair of general type. Then there is a minimal model $f : X \dashrightarrow X_{\min}$ (i.e. $K_{X_{\min}} + f_\Delta$ is semiample).*

Corollary

Flips exist (for KLT pairs; in any dimension).

Remark: We are unable to show termination of flips. But we can pick a sequence of flips that terminates.

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- (Birkar-Cascini-Hacon-M^cKernan) If flips exist in dimension $n + 1$ (and minimal models exist in dimension n) then minimal models exist in dimension $n + 1$. (Uses the ideas of the MMP and in particular techniques developed by Shokurov.)

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- By adjunction, we may write $(K_X + \Delta)|_S = K_S + \Delta_S$ where (S, Δ_S) is KLT.
- Shokurov's observation is that to show that $R(K_X + \Delta)$ is finitely generated over Z (and hence that the flip exists), it suffices to show that the restricted algebra

$$R_S(K_X + \Delta) := \text{Im}(R(K_X + \Delta) \rightarrow R(K_S + \Delta_S))$$

is finitely generated (over Z).

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- Using ideas of Siu, Kawamata and Tsuji we show that $H^0(m(K_S + \Theta_m)) = \text{Im}(H^0(m(K_X + \Delta)) \rightarrow H^0(m(K_S + \Delta_S)))$.

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- The divisor Θ is a priori an $\mathbb{R}$ -divisor.
- Using a Diophantine approximation argument, we show that Θ is in fact a $\mathbb{Q}$ -divisor and $\Theta = \Theta_m$ for some $m > 0$.

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- If $t_1 = 0$ we are done. Otherwise we flip (resp. perform a divisorial contraction) along a ray $R = \mathbb{R}^+[C]$ where $(K_X + \Delta + t_1 A) \cdot C = 0$.
- In this way, we obtain a sequence of rational numbers $1 \geq t_1 \geq t_2 \cdots \geq 0$ and of flips/div. contractions $X_i \dashrightarrow X_{i+1}$ such that X_i is a minimal model for $(X, \Delta + t_i A)$.

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- This implies that the sequence $\dots X_i \dashrightarrow X_{i+1} \dots$ is not infinite and so the above process produces the required minimal model.
- Note that to prove the above result on the finiteness of minimal models, we must use a compactness argument and hence it is critical to work with $\mathbb{R}$ -divisors.